

Aman Sharma

+91 9871987731  GitHub  LinkedIn  amands.5153@gmail.com

SUMMARY

AI engineer with a statistics foundation. I build GenAI applications end-to-end, from RAG pipelines and agentic workflows to fine-tuning and the evaluation rigor that makes them reliable in production.

EXPERIENCE

AI Agent Developer Intern

May 2026 – June 2026

Genotek Global

Remote

- Built and deployed a **Telegram bot** on AWS EC2 for field sales, with GPS-verified check-in/check-out, a 3-gate anti-cheat system, and HubSpot CRM sync; validated end-to-end with company leadership
- Shipped an LLM meeting-debrief pipeline: local **faster-whisper** transcription feeding a **Groq LLaMA 3.1 70B** extraction step, writing structured records into HubSpot, on a 15-state conversation state machine with session isolation and manager alerting

Data Science Intern

Sep 2025 – Nov 2025

Climate Resilience Observatory (CRO), IIIT Lucknow

Lucknow, India

- Analyzed live meteorological data from NASA POWER and ECMWF ERA5 across 72 Uttar Pradesh districts, then ran model selection and evaluation for binary rainfall forecasting with the Office of the Relief Commissioner, Govt. of Uttar Pradesh

PROJECTS

WealthOS — Multi-Agent Financial Intelligence Platform

LangGraph, FastAPI, Qdrant, DSPy, E2B, LangSmith

- Built a **personal AI advisor** that recommends Buy/Hold/Avoid against a user's income, debts, and risk profile: **8 agents** on a LangGraph state machine (45 tools across 7 MCP servers), delivering each memo in 60–90s via asyncio.gather parallelism on Groq llama-3.3-70b, with Pydantic-validated outputs and PII-masked LangSmith traces
- Grounded every recommendation in real SEC filings and computed math: hybrid **Agentic RAG** over 10-K filings (dense + BM25, RRF fusion, Cohere rerank, \$0 embedding cost) with DSPy-compiled prompts and a 5-dim Financial Health Score

Resolv — Guardrail Refund Agent & pass^k Reliability Benchmark

Python, Google ADK, FastAPI, litellm, TRL (GRPO)

- Built a **harness-first refund agent**: two LLM stages (extract, resolve) only propose actions, while 7 first-match policy rules in Python are the sole write path, vetoing every payout before it executes; refund caps are computed from the order record, not the customer's claim, so inflated requests fail deterministic validation rather than model judgment
- Stress-tested against an adversarial AI “customer” across 200 runs: **0 unauthorized refunds**, pass³ 0.96; ablation showed the harness *off* leaked \$2,120 vs \$0 *on*; fine-tuned the extractor via **RLVR/GRPO** (Qwen2.5-1.5B LoRA), lifting accuracy 49%→71%; backed by 240 tests, regression-gated CI, and Docker

RetailGraph — Multimodal Product Knowledge Graph

Qwen2-VL-7B, QLoRA, GraphRAG, Neo4j, Groq, RAGAS

- Built an **AI shopping assistant** over a 2,160-product grocery catalog (e.g. “vegan snacks under \$10”) that combines a knowledge graph with semantic search; fine-tuned **Qwen2-VL-7B** (QLoRA) to read product listings, text and images, and extract structured data at 94.2% accuracy
- Designed a **6-node LangGraph agent** that routes each query to graph traversal, vector search, or aggregation; proved graph retrieval essential, as vector-only search failed every constraint query (0/20), validated on a 30-query eval harness (96.7% route accuracy)

EDUCATION

Indian Institute of Information Technology, Lucknow

2025 – Expected Aug 2027

M.Sc. Data Science | SGPA: 8.65/10

University of Lucknow

2022 – 2025

B.Sc. Statistics

Lucknow, India

SKILLS

Agentic AI: Google ADK, MCP, LangGraph, LangChain, Agentic RAG, GraphRAG, DSPy, Mem0, Policy Engines & Guardrails
LLM & Fine-tuning: SFT, ORPO, GRPO, RLVR, QLoRA, LoRA, LlamaFactory, HuggingFace, LLM-as-Judge, Ragas, Hybrid Search, Prompt Engineering, Vision-Language Models (Qwen2-VL)

Languages: Python (PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib), C, SQL

Mathematics & Statistics: Linear Algebra, Calculus, Probability Theory, Statistical Inference, Hypothesis Testing

Machine Learning: Transformers, Weak Supervision (Snorkel), Supervised & Unsupervised Learning, CNN, LSTM

Databases & Search: Neo4j (Cypher), Qdrant, pgvector, FAISS, Redis, PostgreSQL

Developer Tools: AWS, Docker, FastAPI, Streamlit, LangSmith, W&B, Modal, MLflow

ACHIEVEMENTS

- IIT JAM MS:** Secured All India Rank **219** in IIT JAM 2025 (Mathematical Statistics)
- GATE Statistics:** Secured All India Rank **493** in GATE 2025
- Kaggle:** Notebook Expert